

When the Mines Close: Coal Employment Shocks and Family Health Impacts in Appalachia

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Abstract

This paper investigates how coal mining employment in Appalachia affects health outcomes across different populations within mining-dependent counties. Using the substitution from coal to natural gas in electricity generation as an exogenous shock to coal demand beginning in the late 2000s, we construct a shift-share instrumental variable that exploits county-level variation in baseline coal dependence. We find that mining employment increases deaths of despair among working-age men, driven entirely by drug poisoning deaths, consistent with occupational exposure to prescription opioids. Mining employment also worsens reproductive health. We observe increases in fetal deaths, low birth weights, and preterm births associated with increased mining activity. However, a tradeoff emerges. Mining employment decline harms populations outside the workforce. We observe a decrease in birth rates as well as increases in deaths of despair among children and the elderly, also driven by drug poisonings. Comparing Appalachian to non-Appalachian coal counties reveals that the observed poisoning pathway is unique to Appalachia and not to coal mining itself. Our results suggest that Appalachian coal counties face competing health costs from both mining employment and its decline, with the burden falling on different members of the community.

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1 Introduction

sparities are not simply artifacts of individual characteristics but reflect fundamental differences in local economic structure, industrial composition, and access to opportunity (Kline and Moretti, 2013). Some places thrive while others are left behind, and these patterns persist across decades.

Between 2007 and 2019, coal mining employment in Appalachia began to collapse as the fracking boom enabled natural gas to displace coal. During this time, coal went from generating nearly 50 percent of electricity in the United States to less than 25 percent. For a region already facing geographic isolation, limited economic alternatives, and concentrated poverty, this employment shock struck particularly hard. The same period saw Appalachia become the epicenter of America’s opioid epidemic, with overdose death rates far exceeding national averages. The temporal and geographic overlap of coal’s decline with surging opioid mortality led observers to draw a direct connection, but the causal connection remains ambiguous.

Recent evidence partially challenges this narrative. Metcalf and Wang (2019) show that coal job losses may actually reduce worker mortality from opioid overdoses, as displaced miners avoid the injuries and prescription opioid exposure endemic to underground mining. This finding suggests a distinct mechanism operating in mining communities. Rather than economic despair driving mortality, such as Pierce and Schott, 2020, occupational exposure to prescription opioids may explain elevated death rates among employed miners. In addition to this conundrum, we know that unemployment effects go beyond the worker. This raises the question: if losing hazardous jobs potentially saves workers’ lives, what happens to their families?

In this paper, we examine how coal employment losses affected health outcomes across different groups in Appalachia. Because coal mining is overwhelmingly male-dominated, with men holding over 95 percent of industry jobs (Bureau of Labor Statistics, 2022), we can separately identify effects on men who lost employment, women who experienced partner job

loss, and children and the elderly exposed to household or community economic stress. We focus on deaths of despair (suicide, poisoning from overdoses, and alcohol-related deaths), fetal mortality, and infant health. The male-dominated nature of coal mining allows us to test whether women increase employment in response to partner job loss, as predicted by the added worker effect, or whether limited economic opportunities in coal-dependent regions prevent such adjustments. To assess whether Appalachia’s unique vulnerabilities amplified these health consequences, we compare outcomes in Appalachian coal counties to coal-dependent regions elsewhere in the United States.

Identifying the causal effects of coal employment losses poses a challenge because local economic conditions and health outcomes may be jointly determined. We address this using a shift-share instrument that combines counties’ baseline coal employment shares with national changes in natural gas usage in electricity generation (Borusyak et al., 2025). The fracking boom that began around 2008 dramatically increased natural gas production, enabling it to displace coal in the electricity sector (Brown and Yücel, 2013; Linn and McCormack, 2019; Coglianese et al., 2020). Counties with higher initial dependence on coal mining experienced larger employment declines as this national energy transition unfolded, but the timing and magnitude of the shift from coal to natural gas was driven by national energy markets and technological change rather than local Appalachian conditions. This approach provides plausibly exogenous variation in local labor market shocks, allowing us to estimate the causal effects of employment loss on intergenerational health outcomes.

We first examine the full population. Consistent with the first stage of our identification strategy, declining coal employment shares increase county-level unemployment rates, confirming that the national shift toward natural gas generated local labor market disruptions. We then estimate effects on mortality outcomes. Similar to Metcalf and Wang (2019), we find that declining coal employment reduces opioid overdose deaths among prime-age men. A one percentage point decrease in coal employment reduces opioid mortality by 1.16 deaths per 100,000. We suggest that that this pattern could reflect

reduced occupational exposure to opioids as miners exit a high-injury industry, selective out-migration of healthier workers, or both.

We next examine gender and age differences in the labor market and health responses to coal employment decline. The added worker effect, first formalized by Lundberg (1985), predicts that women increase labor supply when their partners lose employment to compensate for lost household income, a pattern confirmed in recent displacement studies (Halla et al., 2020). We find no evidence of an added worker effect. Rather than increasing labor supply to compensate for partner job loss, women’s employment declines alongside mining employment, suggesting complementarity rather than substitution in local labor markets. Women show no change in overdose deaths, but mining employment worsens reproductive health. Fetal deaths, preterm births, and low birth weight births all increase with mining activity. These outcomes are known to be tied to substance use during pregnancy, Graeve et al. (2022); Giovannini et al. (2024), but warrant further investigation.

The health consequences extend beyond working-age adults. When mining employment declines, overdose deaths increase among children and the elderly, suggesting that economic stability from mining protects vulnerable populations who face neither the occupational hazards of the industry nor the benefit of reduced exposure when jobs disappear. This contrasts with the broader deaths of despair literature showing job displacement increases suicide, and highlights how Appalachia’s opioid crisis has created a distinct pathway through which economic shocks translate into mortality (Case and Deaton, 2020). In-line with the aforementioned mechanisms, the health consequences operate through drug overdoses rather than the suicide and alcohol mechanisms.

Extending the analysis beyond Appalachia highlights what makes the region’s experience unique. Coal employment decline in Appalachian counties drives deaths of despair almost entirely through drug poisonings, with no significant effects on suicide or alcohol-related mortality. Outside Appalachia, mining employment is associated with increased alcohol-related deaths rather than drug poisonings. These results underscore that

Appalachia’s geographic isolation, concentrated poverty, and limited economic alternatives amplify and shape how economic shocks translate into mortality in ways not observed in other coal-producing regions.

A growing body of evidence demonstrates that economic shocks generate severe health and wellbeing consequences, both for displaced workers and their families. Workers who lose jobs to import competition face elevated mortality from overdoses, suicides, and alcohol abuse (Pierce and Schott, 2020), part of a broader pattern of “deaths of despair” concentrated among working-class Americans facing declining economic prospects (Case and Deaton, 2020). Farmers experiencing climate-driven crop failures show increased suicide rates (Carleton and Hsiang, 2017). These health crises extend beyond workers themselves. Timber industry contractions increased domestic violence and child maltreatment as economic stress destabilized families (Lindo et al., 2018), while parental job loss worsened infant health through income shocks and lost healthcare access (Lindo, 2011). This literature establishes that job loss harms both workers and their households, yet it has not examined deaths of despair and opioid mortality in the communities at the center of America’s ongoing opioid epidemic. We focus on coal employment losses in Appalachia during the period when the region faced catastrophic increases in overdose deaths, testing whether mine closures triggered opioid-related mortality and deaths of despair across men, women, and children in affected households.

Work on coal mining in Appalachia has documented the economic devastation of the industry’s decline. Coal boom and bust cycles create asymmetric spillovers, with each mining job lost triggering 3.5 additional job losses elsewhere in the local economy (Black et al., 2005). More recently, coal employment declines have driven population loss and increased reliance on government transfers while failing to generate offsetting human capital investments (Krause, 2023). Selective out-migration of better-educated residents has left coal-dependent communities more vulnerable to subsequent shocks (Krause, 2025). Displaced coal workers face persistent earnings penalties and limited success transitioning to new

industries or regions (Colmer and Krause, 2024). We extend this work by moving beyond economic outcomes to examine health consequences.

Finally, the closest work to ours is that of Metcalf and Wang (2019). They utilize a shift-share instrument and find similar overdose results for working aged men and women. We replicate and extend these results using the more updated shift share literature methodology laid out by Borusyak et al. (2025) and including data through 2019. We go further to investigate and explain the impacts of coal employment beyond workers' deaths. Moreover, by comparing Appalachian coal regions to other U.S. mining areas, we test whether the region's unique combination of geographic isolation, mono-industry dependence, and pre-existing health vulnerabilities made coal job losses particularly deadly, or whether similar employment shocks generated similar health crises regardless of local context.

We do several robustness checks to verify these results. Following Borusyak et al. (2025), we correct for incomplete shares and find no meaningful differences in the results. We also control for the lagged instrument to address serial correlation in the shift component, following Jaeger et al. (2018). Finally, we try to account for differences in population using population-weighted estimates. Our results on poisoning deaths and birth outcomes are successfully replicated in each case.

These results underscore the importance of tailoring policies during an industry collapse to address impacts beyond displaced workers themselves. Children, the elderly, and partners of displaced workers bear significant health consequences that targeted retraining or severance programs for laid-off miners would not reach. Moreover, because economic shocks manifest differently across regions, policy responses designed for one context may be ineffective or misallocated in another.

2 Data

We combine restricted-access mortality data from the National Center for Health Statistics (NCHS) with county-level employment and demographic data to examine how coal employment losses affected deaths of despair and fetal mortality across U.S. counties from 2007 to 2019.

2.1 Data Construction

Our primary analysis sample consists of the 423 counties designated as part of the Appalachian region by the Appalachian Regional Commission (ARC) (Appalachian Regional Commission, 2025). This region spans 13 states from southern New York to northern Mississippi, with particularly heavy concentrations of mining employment in eastern Kentucky, West Virginia, southwestern Virginia, and eastern Tennessee. Figure 1 shows the region with mining counties highlighted in purple. We focus on Appalachia because it contains the vast majority of U.S. mining employment east of the Mississippi River and because the region’s economic and demographic characteristics (high poverty, low educational attainment, geographic isolation, and pre-existing opioid crisis) make it a particularly policy-relevant setting for understanding the health consequences of energy transitions. Our analysis uses a balanced county-year panel consisting of 5,499 observations (423 counties for 13 years).

2.2 Mortality

Our primary outcomes come from the National Vital Statistics System (NVSS) restricted-use mortality files, which we obtained through application to the Centers for Disease Control and Prevention’s National Center for Health Statistics (National Center for Health Statistics, 2019). These data provide complete-count birth and death records for all U.S. counties, disaggregated by sex, age, and cause of death. We use these data to construct annual county-level mortality rates for deaths of despair across multiple demographic groups

including prime working-age men and women (ages 18-64), children and adolescents (ages 0-17), and elderly (65+). Deaths of despair are subdivided into three categories: poisoning deaths (overdoses), suicide, and alcohol-related deaths. From the NVSS natality files, we construct county-level measures of birth rates, mean birth weight, and the incidence of adverse birth outcomes. Low birth weight (LBW) is defined as less than 2,500 grams (World Health Organization, 2014), very low birth weight (VLBW) as less than 1,500 grams (Cutland et al., 2017), and preterm birth as delivery before 37 completed weeks of gestation (World Health Organization, 2012). These measures are constructed from the natality files covering live births, which is a distinct population from the fetal death files described above.

We also examine fetal mortality using the NVSS restricted-use fetal death files. States require reporting of fetal deaths at 20 weeks of gestation or more (or birth weight of 350 grams or more) (Gregory et al., 2025). Following standard NCHS practice, we define fetal deaths as intrauterine deaths at 20 or more weeks of gestation (MacDorman and Gregory, 2015). Fetal deaths exclude induced terminations of pregnancy. These data allow us to test whether coal employment shocks increased pregnancy loss, a potential mechanism linking economic distress and women’s opioid use to reproductive health outcomes according to Whiteman et al. (2014).

2.3 Employment Data

We construct our measure of coal employment shocks using data from the Quarterly Census of Employment and Wages (QCEW), which provides near-universal coverage of wage and salary employment by industry and county (Bureau of Labor Statistics, 2019). We define mining employment as all employment in NAICS code 212 (mining, except oil and gas). The QCEW data have minimal suppression issues at the county-year level for our study period, providing reliable measures of mining employment in even small mining-dependent counties. We use total employment across all industries in a county from the QCEW as the denominator in our exposure measure.

Following the shift-share identification strategy common in the regional labor markets literature (Bartik, 1991; Goldsmith-Pinkham et al., 2020), we construct a Bartik-style instrument that interacts county-level exposure to mining employment with a national-level shock to coal demand. The “share” component is the baseline 2006 mining employment as a fraction of total county employment. The “shift” component captures the exogenous national shock to coal demand driven by the rapid expansion of natural gas production following the hydraulic fracturing boom. Specifically, we use data from the U.S. Energy Information Administration (EIA) on the share of total U.S. electricity generation from natural gas in each year (U.S. Energy Information Administration, 2019). As natural gas displaced coal in electricity generation (rising from 22% of generation in 2007 to 38% in 2019) counties with larger baseline mining shares faced larger employment contractions.

2.4 Opioid Distribution Data

To examine potential supply-side mechanisms, we use data from the Drug Enforcement Administration’s Automation of Reports and Consolidated Orders System (ARCOS), made available through a public-use portal maintained by the Washington Post (University of Notre Dame, 2024). ARCOS tracks every opioid pill distributed in the United States, reported by manufacturers and distributors as required by the Controlled Substances Act. We aggregate transaction-level data to annual county-level measures of morphine milligram equivalents (MME) per capita for oxycodone, hydrocodone, and other commonly prescribed opioids. These data allow us to distinguish whether opioid mortality responds to labor market shocks through increased pharmaceutical supply or through demand-side mechanisms such as pain, despair, or self-medication.

2.5 Demographic Controls

We use American Community Survey (ACS) 3-year and 5-year estimates to construct annual county-level demographic controls including median age, racial composition, median

household income, and high school attainment (share with bachelor's degree or higher) (U.S. Census Bureau, 2019). For counties with populations below the ACS 3-year publication threshold (typically counties under 65,000 population), we use 5-year estimates centered on the survey year. For 2007-2008, when ACS coverage is limited to counties with population over 25,000, we use intercensus population estimates to fill remaining gaps. These controls allow us to account for systematic differences in population composition that might confound the relationship between mining employment shocks and mortality outcomes.

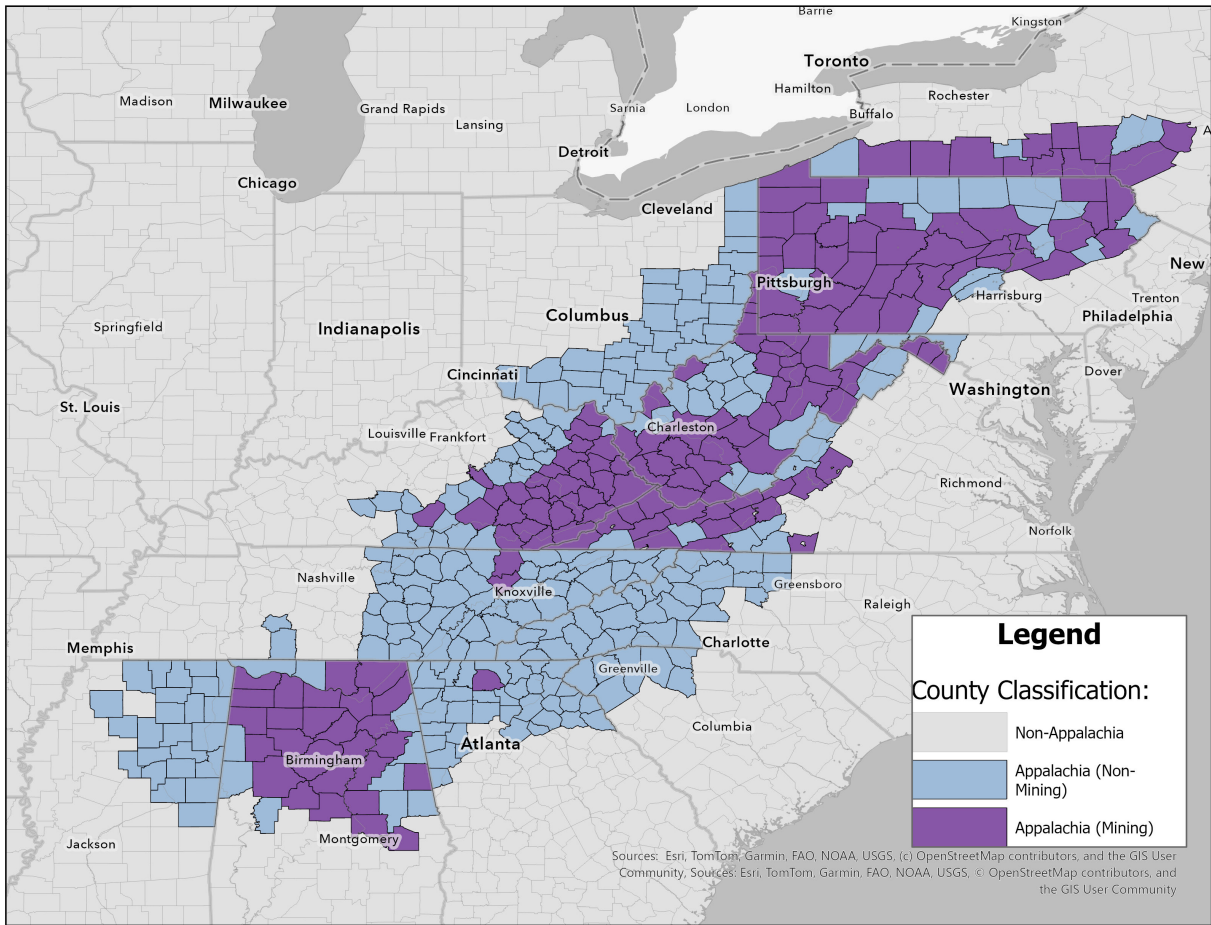


Figure 1: Appalachia Region Classification

For our geographic comparison analysis, we extend the sample to include all U.S. counties. Non-Appalachian coal counties are primarily in Wyoming, Montana, Illinois, and other western and Midwestern states. This comparison allows us to test whether Appalachia's

unique vulnerabilities amplified the health consequences of coal employment losses relative to other coal-dependent regions.

2.6 Summary Statistics

Table 1 shows the summary statistics for the sample of high, low, and no mining counties. The average share of mining employment in high mining counties is 11.9 percent. Counties in Appalachia are overwhelmingly white, over 90% across groups. Unemployment and labor force participation rates are similar in each group. High mining counties are small on average, but have similar birth rates to other counties in the region.

Table 1: Summary Statistics by Mining Classification, Appalachian Counties, 2009

	No Mining ($N = 324$)		Low Mining ($N = 68$)		High Mining ($N = 31$)	
	Mean	SD	Mean	SD	Mean	SD
<i>Panel A: Demographics</i>						
Total Population	47,236	69,067	123,269	168,414	38,506	31,258
Percent White	0.903	0.117	0.910	0.115	0.936	0.081
Percent Black	0.068	0.113	0.055	0.111	0.047	0.078
Percent Hispanic	0.022	0.026	0.030	0.050	0.010	0.010
Percent Bachelor's+	14.60	6.23	18.29	7.35	11.96	4.31
Median Household Income	36,865	7,463	41,112	9,518	33,887	6,284
<i>Panel B: Labor Market</i>						
Mining Employment Share (%)	0.09	1.21	0.44	0.55	11.19	10.42
Labor Force Participation	45.51	4.68	47.30	3.96	40.65	5.61
Unemployment Rate	8.07	2.72	7.74	2.67	7.97	2.25
Poverty Rate	18.21	5.79	15.93	5.43	21.24	6.24
<i>Panel C: Health Outcomes</i>						
Deaths of Despair Rate	38.47	17.55	31.57	12.15	42.88	17.40
Poisoning Rate	17.01	11.77	14.20	8.55	21.92	16.44
Suicide Rate	16.50	9.74	13.20	6.14	15.49	7.93
Alcohol Rate	6.20	5.81	5.52	3.82	6.09	5.38
Fetal Death Rate	6.78	6.46	7.31	4.61	8.51	6.19
Birth Rate	11.49	2.06	11.53	2.10	11.82	1.47
Opioid Rx Rate	61,795	32,406	60,354	25,685	84,702	45,188

Notes: Appalachian counties in 2009 classified by 2006 baseline mining employment share: No Mining (0%), Low Mining (0–2%), High Mining ($\geq 2\%$). Death rates per 100,000 total population. Birth rate per 1,000 total population. Opioid prescription rate per 100,000 total population in grams of active ingredient distributed (ARCOS). Mining employment share reported in percentage points after rescaling.

3 Estimation Strategy

Beginning in the late 2000s, technological advances in hydraulic fracturing and horizontal drilling triggered a rapid increase in natural gas production, which displaced coal in electricity generation markets. Figure 2 shows natural gas shares of total electricity generation nearly doubling from 2007 to 2019. This national energy market shift generated differential employment shocks across Appalachian counties based on their baseline dependence on mining.

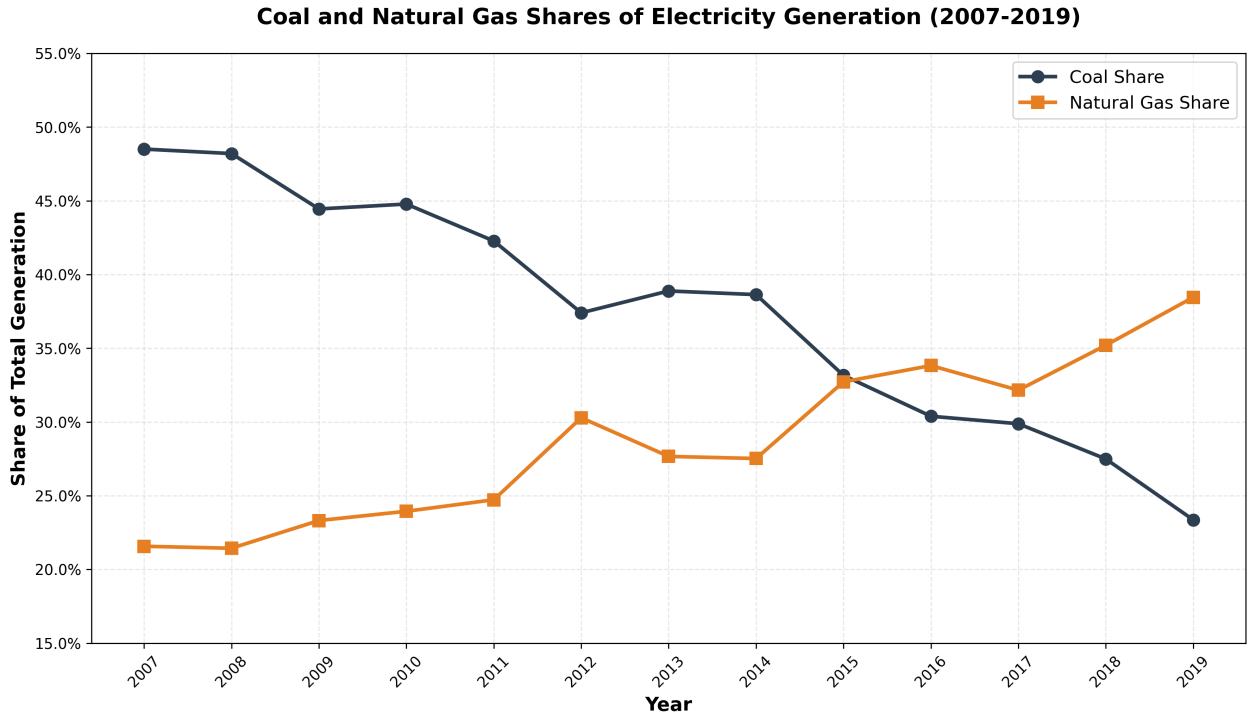


Figure 2: Coal and Natural Gas Shares of Total Net Generation, 2007-2019

3.1 Baseline Specification and Challenges

We begin by estimating the relationship between mining employment shares and health outcomes using a simple two-way fixed effects specification:

$$Y_{c,s,t} = \beta_1 \text{EmpShare}_{c,t} + \gamma X_{c,t} + \alpha_c + \delta_t + \eta_{s,t} + \epsilon_{c,s,t} \quad (1)$$

where $Y_{c,s,t}$ is the mortality outcome (deaths of despair, fetal deaths, or births) in county c , state s , and year t . $\text{EmpShare}_{c,t}$ is county c 's mining employment share in year t , defined as mining employment (NAICS 212) divided by total county employment. $X_{c,t}$ includes time-varying county demographic controls from the ACS. α_c are county fixed effects, δ_t are year fixed effects, and $\eta_{s,t}$ are state-by-year fixed effects.

The baseline specification faces a key identification challenge. Time-varying factors may jointly determine both mining employment and health outcomes in these counties. For example, supply-side shocks to the opioid epidemic (such as increased pharmaceutical marketing or pill mill proliferation) could simultaneously reduce employment through workforce disability and directly increase overdose deaths. Similarly, broader regional economic decline or changes in healthcare infrastructure could confound the relationship between mining employment and mortality. To address these threats to identification, we implement a shift-share instrumental variables design that isolates variation in mining employment driven by national energy market shifts.

3.2 Shift-Share Instrument Construction

We implement this identification strategy using a shift-share instrumental variables design (Bartik, 1991). The shift-share approach isolates quasi-experimental variation by interacting a common aggregate shock (the “shift”) with heterogeneous local exposure to that shock (the “share”). Following the exogenous shifts framework of Borusyak et al. (2025), we treat the national shift toward natural gas as an exogenous shock and exploit cross-sectional variation in county-level exposure to this shock through their differential initial shares of coal mining employment.

The shift-share instrument takes the general form:

$$z_{ct} = \sum_{k=1}^K s_{ck} \cdot g_{kt} \quad (2)$$

In our setting, there is a single shift dimension, the coal industry. Thus $K = 1$, and the summation collapses to:

$$z_{ct} = s_c \times g_{t-1} \quad (3)$$

where s_c is the 2006 share of mining employment in county c (measured as mining employment divided by total county employment) and g_{t-1} is the lagged national ratio of natural gas to coal electricity generation. We use the lagged ratio rather than the contemporaneous value to account for the delayed response of local employment to national energy market shifts. Figure 3 shows the geographic distribution of mining employment shares across Appalachian counties in 2006.

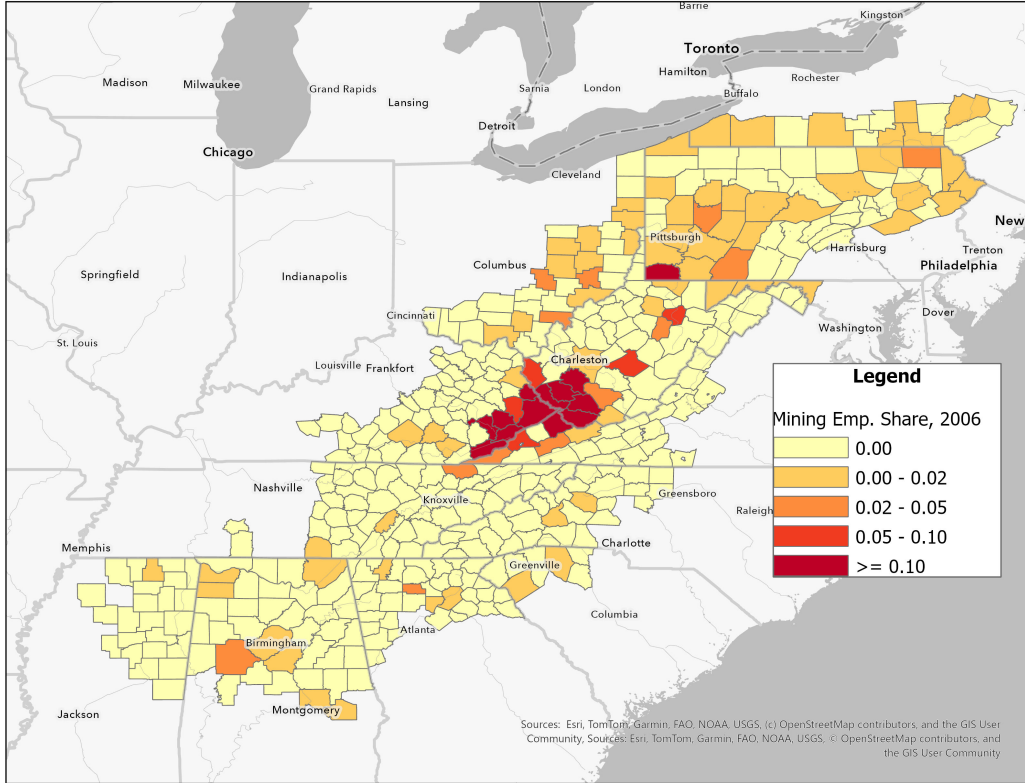


Figure 3: Mining Employment Shares - 2006

Our design differs from the canonical cross-sectional shift-share instrument in an important way. We exploit time-series variation in a common national shift rather than cross-sectional variation across multiple industry-specific or location-specific shifts. We

observe a single aggregate shift per period, the national gas-to-coal ratio, that varies over time and affects all counties simultaneously but differentially based on their initial mining dependence. This panel structure, similar to Nunn and Qian (2014), creates multiple quasi-experiments. Counties with high mining dependence experience larger employment shocks in years when natural gas displaces more coal nationally. Identification comes from the interaction of cross-sectional differences in exposure with the timing and magnitude of the aggregate energy market shock.

The key identifying assumption is that the national shift toward natural gas is orthogonal to unobserved determinants of county-level mortality trends. This assumption is plausible because the fracking boom was driven by exogenous technological breakthroughs and geological discoveries of shale deposits, rather than conditions in Appalachian labor or health markets. Moreover, individual coal-producing counties are too small to influence national electricity generation patterns. Formally, our instrument must satisfy two conditions: (1) *Relevance*: $\text{cov}(z_{ct}, \text{EmpShare}_{ct}) \neq 0$, and (2) *Exogeneity*: $\text{cov}(z_{ct}, \epsilon_{cst}) = 0$. Our single-shift design raises an incomplete shares concern (Borusyak et al., 2025), which we address in Section 5 using baseline county characteristics interacted with year fixed effects.

3.3 Instrumental Variables Specification

We estimate the effect of mining employment shares on deaths of despair using a two-stage least squares approach. The second stage estimates:

$$Y_{c,s,t} = \beta_1 \widehat{\text{EmpShare}}_{c,t} + \gamma X_{c,t} + \alpha_c + \delta_t + \eta_{s,t} + \epsilon_{c,s,t} \quad (4)$$

where $Y_{c,s,t}$ is the mortality outcome of interest per 100,000 population in county c , state s , and year t . $\widehat{\text{EmpShare}}_{c,t}$ represents the predicted mining employment share from the first stage. $X_{c,t}$ is a vector of time-varying county demographic controls from the ACS. α_c are county fixed effects, δ_t are year fixed effects, and $\eta_{s,t}$ are state-by-year fixed effects.

The first stage relates county-level mining employment shares to our shift-share

instrument:

$$\text{EmpShare}_{c,t} = \psi z_{c,t} + \gamma X_{c,t} + \alpha_c + \delta_t + \eta_{s,t} + \nu_{c,s,t} \quad (5)$$

where $z_{c,t}$ is our shift-share instrument defined in equation 3¹. Identification comes from within-county variation such that counties with higher initial mining shares experience larger employment declines when the national natural gas to coal ratio rises. The county fixed effects absorb permanent differences in mortality levels across counties, while year fixed effects absorb national trends in deaths of despair during the opioid epidemic.

The state-by-year fixed effects control for state-specific policies and economic conditions in each year, including variation in Medicaid expansion, opioid regulations, and state-level labor market trends. This specification leverages within-state variation in mining dependence while absorbing systematic differences in how states responded to the opioid epidemic and economic shocks during this period. These fixed effects address the primary exclusion restriction concern that state-level energy prices or opioid policies could correlate with both the instrument and mortality outcomes, ensuring identification comes from within-state variation in baseline coal dependence.

Outcomes include total deaths of despair, disaggregated into drug poisoning deaths (overdoses), suicides, and alcohol-related deaths per 100,000 people. We also examine fetal mortality to test whether mining employment shocks increased pregnancy loss. We estimate these effects for the overall population and separately by gender and age group.

4 Results

Before presenting our causal estimates, we first document the time-series variation in the national energy market shift and the geographic distribution of mortality changes across Appalachian counties. Figure 4 shows the proportion of natural gas to coal in electricity

¹For ease of interpretation, both the mining employment share and the shift-share instrument are rescaled by a factor of 100.

generation over the sample period. A clear pattern emerges. The ratio rises steadily from 0.45 in 2007 to 1.68 in 2019, crossing parity in 2015. Figure 2 shows these trends independently over time. This shift of energy generation toward natural gas establishes the national demand shock that differentially affected counties based on their initial dependence on mining employment.

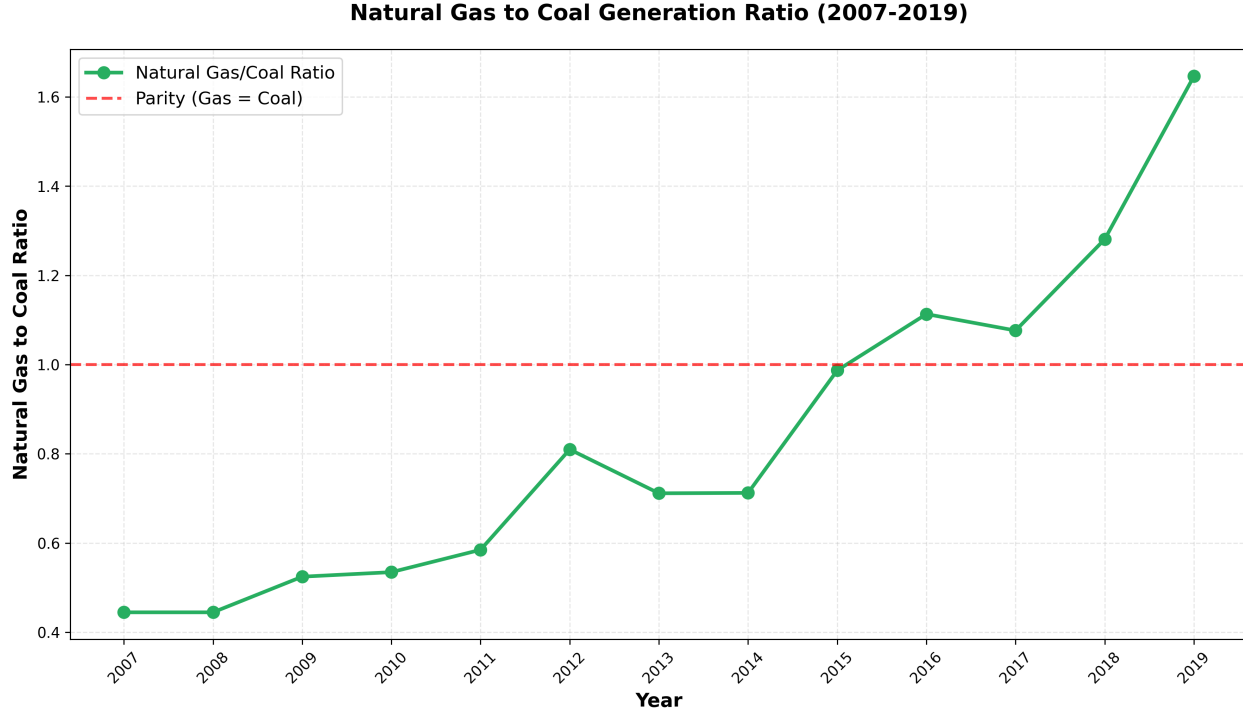
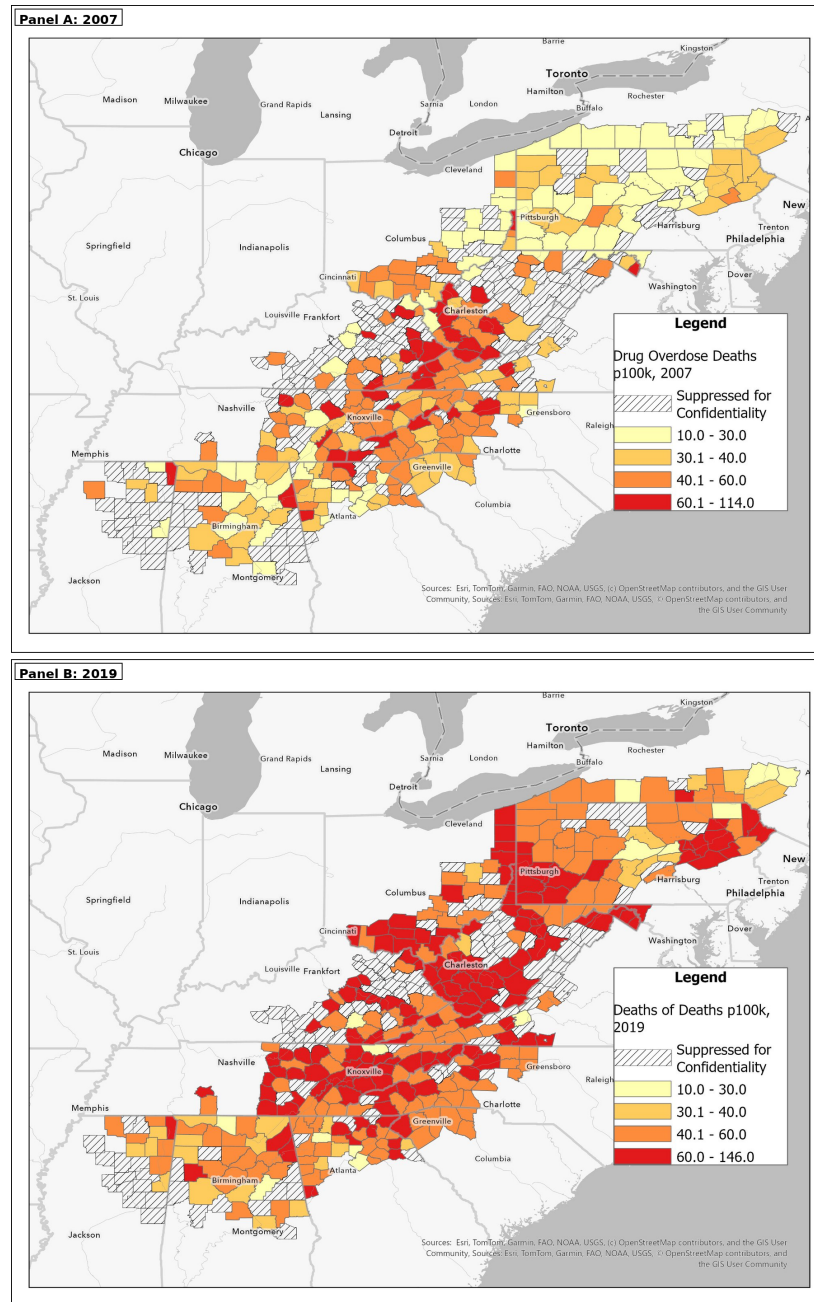


Figure 4: Natural Gas to Coal Generation Ratio, 2007-2019

Figure 5 shows the variation in deaths of despair over time across Appalachian counties. The maps reveal substantial geographic heterogeneity, with the heart of coal country in eastern Kentucky and southern West Virginia experiencing increases in deaths of despair exceeding 50 percent, while some southern Appalachian counties saw modest declines or stable mortality rates over the same period. This geographic variation in mortality trends motivates our empirical analysis of whether mining employment shocks contributed to these divergent outcomes across Appalachian counties.



Note: Suppressed counties are due to the fact that NCHS does not allow for visual representation for counties where fewer than 10 deaths of despair occurred in a given year.

Figure 5: Deaths of Despair in Appalachia, 2009 and 2019

4.1 First Stage

Table 2 presents the first-stage relationship between our shift-share instrument and county-level mining employment shares. The coefficient on the interaction between the

lagged natural gas to coal ratio and baseline (2006) mining employment share is negative and highly statistically significant across all specifications. The estimate is stable, ranging from -0.79 to -0.80 across columns (1)-(3), indicating that counties with higher initial mining dependence experienced larger employment share declines as natural gas displaced coal in electricity generation.

The magnitude implies that a one-unit increase in the gas-to-coal ratio (roughly the change from 2007 to 2019) reduced the mining employment share by approximately 0.8 percentage points in a county at the sample mean of baseline mining dependence. Given that the mean mining employment share in 2006 was 0.81 percent, this represents a substantial proportional decline. The instrument remains strong across specifications. Column (1) includes county and year fixed effects; column (2) replaces year fixed effects with state-by-year fixed effects to control for state-specific policies and economic conditions; column (3) adds time-varying demographic controls. The F-statistics far exceed conventional thresholds for weak instruments, confirming the relevance of our shift-share design.

Table 2: First Stage: Mining Employment Response to Energy Shifts

	Mining Employment Share		
	(1)	(2)	(3)
Lag Natural Gas to Coal Ratio $\times$ 2006 Mining Emp. Share	-0.7912^{***} (0.0875)	-0.7957^{***} (0.0862)	-0.7964^{***} (0.0861)
Mean Dep. Var.	0.8066	0.8066	0.8066
F-statistic	81.76	85.03	85.34
Observations	5,499	5,499	5,499
County FEs	Yes	Yes	Yes
Year FEs	Yes	No	Yes
State $\times$ Year FEs	No	Yes	Yes
Demographic Controls	No	No	Yes

Notes: Mining employment share is measured in percentage points. Demographic controls include percent white, percent Black, percent Hispanic, and percent aged 18–64. Standard errors clustered by county in parentheses. *** $p < 0.01$.

4.2 Main Results

Table 3 presents our main estimates of the relationship between mining employment and deaths of despair. Panel A shows OLS estimates from the baseline two-way fixed effects specification. A one percentage point increase in mining employment share is associated with a 0.58 per 100,000 increase in total deaths of despair (column 1), driven primarily by drug poisoning deaths (0.53 per 100,000 in column 2). We find no significant relationship with suicide or alcohol-related deaths in any specification.

Panel B presents reduced-form estimates, regressing outcomes directly on the shift-share instrument. These estimates capture the direct effect of exposure to the national energy market shift on mortality, without scaling by the first-stage employment response. Counties more exposed to coal displacement (higher baseline mining shares interacted with rising gas-to-coal ratios) experienced significant reductions in deaths of despair (0.93 per 100,000) and drug poisoning deaths (0.84 per 100,000).

Panel C reports our preferred two-stage least squares estimates, which scale the reduced-form effects by the first-stage relationship to recover the causal effect of mining employment changes. The IV estimates indicate that a one percentage point decline in mining employment share causes deaths of despair to fall by 1.16 per 100,000 (column 1). This effect is entirely driven by drug poisoning deaths, which decline by 1.06 per 100,000 (column 2).^{2 3}

These results reveal a tension, declining mining employment reduces deaths of despair rather than increasing them. This finding contradicts the conventional narrative linking job loss to mortality increases, as documented by Pierce and Schott (2020) for manufacturing displacement and Case and Deaton (2020) for broader economic decline.

To understand this result, it is crucial to recognize that mining employment and mining-related deaths move in the same direction. When mining employment rises,

²Appendix Table A.1 shows Panel C over alternative specifications.

³Appendix Tables A.2 and A.3 verify these results using death counts and population weights to account for differences in population.

overdose deaths rise; when mining employment falls, overdose deaths fall. The first-stage F-statistics exceed 85 across all specifications, confirming that our instrument strongly predicts employment changes. Relative to the sample mean death rate of 45.6 per 100,000, a one percentage point decline in mining employment share would reduce deaths of despair by approximately 2.5 percent.

This pattern is consistent with mining employment creating workplace injuries leading to opioid prescriptions, or through the physical demands and hazards of mining work, rather than job loss causing despair-related mortality. These results suggest that mining employment itself may be a pathway to fatal overdoses. The heterogeneity analysis that follows examines whether this aggregate effect masks important differences across demographic groups.

Table 3: Deaths of Despair: OLS and IV Estimates

	(1) Despair	(2) Poisoning	(3) Suicide	(4) Alcohol
<i>Panel A: OLS</i>				
Mining Employment Share	0.5811*** (0.2084)	0.5272** (0.2213)	0.0383 (0.0644)	0.0049 (0.0478)
<i>Panel B: Reduced Form</i>				
Instrument (z_{it})	-0.9269*** (0.2945)	-0.8440*** (0.2791)	0.0018 (0.0912)	-0.0816 (0.0660)
<i>Panel C: 2SLS</i>				
Mining Employment Share	1.1639*** (0.4368)	1.0598*** (0.4081)	-0.0023 (0.1145)	0.1024 (0.0827)
Mean Dep. Var.	45.5904	21.6477	17.4506	7.9223
First-stage F-stat	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes

Notes: Rates per 100,000 total population. For ease of interpretation, both the mining employment share and the shift-share instrument are rescaled by a factor of 100. Therefore, Panel A and C coefficients represent the effect of a one percentage point increase in mining employment share. Panel B coefficients report the effect of a one-unit increase in the shift-share instrument on the outcome directly. Demographic controls include percent white, percent Black, percent Hispanic, and percent aged 18–64. Standard errors clustered by county in parentheses.
* p<0.10, ** p<0.05, *** p<0.01.

4.3 Heterogeneity: Deaths of Despair by Gender and Age

Table 4 disaggregates the main results by gender, revealing that the aggregate protective effect of mining employment declines masks stark heterogeneity between men and women. Panel A shows that for men, a one percentage point increase in mining employment share increases deaths of despair by 2.26 per 100,000 male population (column 1), driven entirely by drug poisoning deaths at 1.95 per 100,000 (column 2). We find no significant effects on male suicide or alcohol-related deaths. In contrast, Panel B shows precisely estimated null effects for women across all causes of death.

These gender-specific patterns are consistent with mining being an occupational pathway to opioid addiction for male workers. Mining is an overwhelmingly male industry as 95 percent of coal miners are men (Bureau of Labor Statistics, 2022) and the physically demanding and injury-prone nature of mining work leads to high rates of opioid prescriptions for pain management. When mining employment rises, male workers face increased exposure to workplace injuries and subsequent prescription opioid use, which can progress to fatal overdoses. Women, who are largely excluded from mining jobs, experience no change in overdose mortality when local mining employment fluctuates.

The magnitudes are economically significant. At the mean male death rate of 66.0 per 100,000, a one percentage point increase in mining employment share raises male deaths of despair by 3.4 percent. The effect on male drug poisoning deaths (1.95 per 100,000) represents a 7.4 percent increase relative to the mean of 26.5 per 100,000. Conversely, the mining employment decline observed during our sample period reduced male overdose deaths, even as other economic pressures from job loss may have increased distress.

Tables 5 and 6 disaggregate these effects by age group. Table 5 shows that mining employment increases deaths of despair most strongly for prime working-age adults, with the largest effects concentrated among those aged 25-34 (0.88 per 100,000) and 35-44 (0.48 per 100,000). In sharp contrast, we find a protective effect for those aged 65 and older. A one percentage point increase in mining employment reduces deaths of despair among the elderly

Table 4: Deaths of Despair by Gender

	(1) Despair	(2) Poisoning	(3) Suicide	(4) Alcohol
<i>Panel A: Male</i>				
Mining Employment Share	2.2649*** (0.8271)	1.9598** (0.8048)	0.1292 (0.2785)	0.2005 (0.1362)
<i>Panel B: Female</i>				
Mining Employment Share	0.0756 (0.2173)	0.1536 (0.3063)	-0.1313 (0.1303)	0.0145 (0.0890)
Mean Dep. Var. (Male)	66.6033	26.5065	28.7187	12.6677
Mean Dep. Var. (Female)	25.0725	16.9128	6.4314	3.2960
First-stage F-stat	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Rates per 100,000 gender-specific population. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

by 0.15 per 100,000 although the effect is statistically insignificant.

Table 6 demonstrates that these age-specific patterns are driven entirely by drug poisoning deaths. The primary effects remain concentrated in prime working ages (25-54). Among the elderly, the negative coefficient of -0.58 per 100,000 indicates that mining employment is protective, and its decline leads to increased drug poisoning deaths in this age group. Among children and adolescents, the negative coefficient (-0.06 per 100,000) indicates that mining employment is similarly protective, with poisoning deaths increasing when mining employment declines. Further, Appendix Tables A.4 and A.5 show that suicide and alcohol related deaths are not more likely to occur in the oldest and youngest groups.

The contrasting effects across age groups reveal that mining employment has competing effects on different populations. For prime-age adults, mining jobs create direct occupational exposure to workplace injuries and prescription opioid pathways, increasing their overdose risk. However, when mining employment declines and communities experience economic hardship, vulnerable populations outside the workforce suffer. Children and the elderly face increased overdose deaths as households lose income and economic despair spreads beyond

the workers themselves.

Table 5: Deaths of Despair by Age Group

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	0–17	18–24	25–34	35–44	45–54	55–64	65+
Mining Employment Share	−0.0604 (0.0943)	0.2117* (0.1103)	0.8790*** (0.2910)	0.4750* (0.2867)	0.3509 (0.2166)	−0.0837 (0.1489)	−0.1525 (0.5443)
Mean Dep. Var.	2.6041	4.5224	12.0043	15.2168	18.5018	13.5060	35.2131
First-stage F-stat	85.56	85.56	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Rates per 100,000 age group-specific population. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Standard errors clustered by county in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Poisoning Deaths by Age Group

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	0–17	18–24	25–34	35–44	45–54	55–64	65+
Mining Employment Share	−0.0625* (0.0320)	0.1984*** (0.0669)	0.8828*** (0.2726)	0.5835** (0.2917)	0.2393 (0.1740)	−0.1351 (0.1307)	−0.5826*** (0.1719)
Mean Dep. Var.	0.5685	2.2943	7.7799	9.4904	9.5580	4.5677	5.2444
First-stage F-stat	85.56	85.56	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Rates per 100,000 age group-specific population. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Standard errors clustered by county in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

4.4 Secondary Results: Birth and Fetal Outcomes

Beyond mortality, mining employment declines also affect reproductive health outcomes. Table 7 examines birth outcomes. Panel C shows that a one percentage point increase in mining employment share raises the birth rate by 0.82 per 1,000 population. This implies that mining employment declines during our sample period reduced birth rates in coal-dependent counties, consistent with economic hardship reducing fertility. However, mining employment also worsens birth outcomes. It significantly increases low birth weight births (1.02 per 100,000) and preterm births (1.51 per 100,000).

Table 8 examines fetal mortality. Panel C shows that mining employment increases total fetal deaths by 0.20 per 100,000 population. This effect is concentrated in late fetal deaths (28+ weeks gestation) at 0.12 per 100,000, while early fetal deaths (20-27 weeks) is statistically insignificant. The increase in fetal deaths is driven by low birth weight fetal deaths (0.16 per 100,000) and very low birth weight fetal deaths (0.14 per 100,000), suggesting that mining employment may worsen maternal health conditions or increase substance use during pregnancy.⁴

Taken together, these reproductive health results reveal another dimension of mining employment's health effects. Higher mining employment increases birth rates but also increases fetal mortality (particularly low birth weight fetal deaths) and worsens birth outcomes among live births (low birth weight and preterm births). This pattern is consistent with the mechanism suggested by Whiteman et al. (2014) linking maternal substance use during pregnancy to adverse reproductive outcomes. When mining employment declines, birth rates fall as economic conditions worsen, but fetal mortality and adverse birth outcomes also decline.

⁴Appendix Table A.6 further looks at heterogeneity in mother's age and Appendix Table A.7 shows an alternative specification using only the female population in the denominator.

Table 7: Birth Outcomes

	(1)	(2)	(3)	(4)	(5)
	Birth Rate	Mean BW	LBW Rate	VLBW Rate	Preterm Rate
<i>Panel A: OLS</i>					
Mining Employment Share	0.0421*** (0.0121)	−0.0740 (0.6923)	0.4661** (0.2177)	0.1285 (0.0816)	1.1664** (0.4954)
<i>Panel B: Reduced Form</i>					
Instrument (z_{it})	−0.0655*** (0.0110)	−0.0226 (0.8371)	−0.8147** (0.3320)	−0.0566 (0.0750)	−1.2001* (0.6640)
<i>Panel C: 2SLS</i>					
Mining Employment Share	0.0822*** (0.0139)	0.0284 (1.0509)	1.0229** (0.4412)	0.0710 (0.0983)	1.5069* (0.8676)
Mean Dep. Var.	10.9535	3,244.75	98.2733	16.1954	142.4690
First-stage F-stat	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes
State × Year FEs	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes

Notes: Panel A estimated by OLS. Panel B reports the reduced form effect of a one-unit increase in the rescaled shift-share instrument on the outcome directly. Panel C estimated by 2SLS. Panels A and C coefficients represent the effect of a one percentage point increase in mining employment share. Birth rate per 1,000 total population. Mean birthweight in grams. LBW defined as <2,500g (World Health Organization, 2014), VLBW as <1,500g (MacDorman and Gregory, 2015), and preterm as <37 weeks gestation (World Health Organization, 2012). Rates per 100,000 total population. Standard errors clustered by county in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 8: Fetal Deaths: Timing and Weights

	(1) Total	(2) Early	(3) Late	(4) Low BW	(5) Very Low BW
<i>Panel A: OLS</i>					
Mining Employment Share	0.1471** (0.0619)	0.0538* (0.0297)	0.0899* (0.0462)	0.1019* (0.0614)	0.0694 (0.0541)
<i>Panel B: Reduced Form</i>					
Instrument (z_{it})	-0.1591** (0.0684)	-0.0648 (0.0506)	-0.0986** (0.0477)	-0.1285** (0.0621)	-0.1115* (0.0671)
<i>Panel C: 2SLS</i>					
Mining Employment Share	0.1999** (0.0859)	0.0814 (0.0665)	0.1238** (0.0515)	0.1613** (0.0738)	0.1400* (0.0825)
Mean Dep. Var.	7.1265	3.2015	3.8783	5.2151	4.0355
First-stage F-stat	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes

Notes: Panel A estimated by OLS. Panel B reports the reduced form effect of a one-unit increase in the rescaled shift-share instrument on the outcome directly. Panel C estimated by 2SLS. Panels A and C coefficients represent the effect of a one percentage point increase in mining employment share. Fetal death rates per 100,000 total population. Early fetal deaths occur at 20–27 weeks gestation; late at 28+ weeks (Gregory et al., 2025). Low birthweight defined as <2,500g (World Health Organization, 2014); very low birthweight as <1,500g (MacDorman and Gregory, 2015). Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Mechanisms: Labor Markets and Opioid Supply

We examine two potential mechanisms through which mining employment affects mortality: labor market conditions and opioid supply. Table 9 confirms that our shift-share instrument generates the expected labor market effects. Panel C shows that a one percentage point increase in mining employment share reduces unemployment by 0.26 percentage points and reduces poverty by 0.26 percentage points. Notably, we find a significant increase in female employment (0.51 percentage points), indicating that mining employment generates spillover effects beyond direct mining jobs, potentially through increased demand for services in mining-dependent communities.

Table 10 examines whether mining employment affects the supply of prescription opioids using ARCOS data on pharmaceutical distribution. We find that mining employment significantly increases hydrocodone distribution (1,305.8 grams per 100,000 population), but

has no significant effect on total opioids or other specific opioid types including oxycodone, fentanyl, morphine, or methadone. Hydrocodone is one of the most commonly prescribed opioids for pain management. The increase in hydrocodone supply is consistent with the occupational injury pathway suggesting that mining employment generates workplace injuries that lead to legitimate pain prescriptions, which can progress to dependence and fatal overdoses.

The combination of these mechanism results clarifies the pathways through which mining employment affects mortality. Mining employment delivers clear economic benefits, reducing unemployment and poverty while increasing female employment through local economic spillovers. However, it simultaneously increases hydrocodone distribution, consistent with workplace injuries generating opioid prescriptions. The absence of effects on other opioid types or total prescription volume suggests a specific occupational injury channel rather than broad pharmaceutical oversupply. Together, these findings support the interpretation that mining jobs create direct health risks for workers through injury-related opioid exposure, even as they improve household economic conditions.

Table 9: Coal Mining Shares and Economic Outcomes

	(1) Unemployment	(2) Poverty	(3) Female Emp.
<i>Panel A: OLS</i>			
Mining Employment Share	−0.1221*** (0.0321)	−0.0956* (0.0549)	0.1353* (0.0792)
<i>Panel B: Reduced Form</i>			
Instrument (z_{it})	0.2050*** (0.0469)	0.2010*** (0.0663)	−0.4118*** (0.1024)
<i>Panel C: 2SLS</i>			
Mining Employment Share	−0.2632*** (0.0695)	−0.2628*** (0.0960)	0.5170*** (0.1633)
Mean Dep. Var.	8.4044	18.4784	36.0108
First-stage F-stat	85.56	85.56	85.56
Observations	5,139	5,245	5,499
County FEs	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes

Notes: All outcomes measured in percentage points. Panel A estimated by OLS. Panel B reports the reduced form effect of a one-unit increase in the rescaled shift-share instrument on the outcome directly. Panel C estimated by 2SLS; Panels A and C coefficients represent the effect of a one percentage point increase in mining employment share. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Coal Mining Shares and Opioid Prescription Distribution

	(1) Total	(2) Hydrocodone	(3) Oxycodone	(4) Fentanyl	(5) Morphine	(6) Methadone
Mining Employment Share	587.1522 (684.2356)	1,305.8066* (778.8212)	201.2519 (277.3124)	−0.9814 (1.5826)	−18.9615 (43.6643)	−26.6506 (23.3406)
Mean Dep. Var.	61,540.69	18,383.98	24,125.10	157.21	7,603.85	1,829.58
First-stage F-stat	85.56	85.56	85.56	85.56	85.56	85.56
Observations	5,115	5,115	5,115	5,115	5,115	5,115
County FEs	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Rates per 100,000 total population measured in grams of active ingredient distributed (ARCOS data). Mining employment share measured in percentage points. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.6 Geographic Heterogeneity: Appalachia vs. Non-Appalachia

Table 11 compares the effects of mining employment on deaths of despair between Appalachian and non-Appalachian coal counties. Columns (1) through (4) replicate our main

results for the Appalachian sample, showing that mining employment significantly increases total deaths of despair (1.16 per 100,000) and drug poisoning deaths (1.06 per 100,000), with no effects on suicide or alcohol-related deaths.

Columns (5) through (8) present results for non-Appalachian coal counties. The pattern differs markedly from Appalachia. We find no significant effects on total deaths of despair or drug poisoning deaths, but observe a significant increase in alcohol-related deaths (1.05 per 100,000, $p < 0.05$). This suggests that the occupational pathway from mining employment to drug overdoses may be specific to the Appalachian context, possibly reflecting regional differences in opioid prescribing practices, healthcare access, or the prevalence of occupational injuries in Appalachian versus Western coal mining.

The contrast between regions indicates that mining employment's health effects vary by geographic and institutional context. The drug poisoning pathway appears strongest in Appalachia, while alcohol-related mortality responds to mining employment outside Appalachia.

Table 11: Deaths of Despair: Appalachia vs. Non-Appalachia

	Appalachia				Non-Appalachia			
	(1) Despair	(2) Poisoning	(3) Suicide	(4) Alcohol	(5) Despair	(6) Poisoning	(7) Suicide	(8) Alcohol
Mining Employment Share	1.1639*** (0.4368)	1.0598*** (0.4081)	-0.0023 (0.1145)	0.1024 (0.0827)	1.4544 (0.9641)	0.0853 (0.5301)	0.0700 (0.6351)	1.0511** (0.4738)
Mean Dep. Var.	45.5904	21.6477	17.4506	7.9223	37.6970	12.6635	16.5216	10.0287
First-stage F-stat	85.56	85.56	85.56	85.56	12.04	12.04	12.04	12.04
Observations	5,499	5,499	5,499	5,499	34,862	34,862	34,862	34,862
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Rates per 100,000 total population. Mining employment share measured in percentage points. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Robustness

5.1 Incomplete Shares Correction

The shift-share instrument uses incomplete shares, as mining employment constitutes only a small fraction of total county employment. Borusyak et al. (2025) show that with incomplete shares, the standard correction of controlling for baseline shares interacted with year fixed effects is collinear with a single-shift instrument. We therefore follow their recommended alternative of interacting observable correlates of baseline mining shares (2009 county unemployment rate, poverty rate, and percent white) with year fixed effects. This allows counties with characteristics typical of high-mining-share areas to follow flexible outcome trends, addressing the concern that our instrument captures differential trends correlated with coal dependence rather than the causal effect of mining employment changes.

Table 12 compares our main specification (Columns 1-4) to the incomplete shares correction (Columns 5-8). The corrected estimates are modestly smaller but remain statistically significant across all outcomes. Deaths of despair decline from 1.16 to 0.93 per 100,000 ($p < 0.05$), poisoning deaths from 1.06 to 0.87 per 100,000 ($p < 0.05$), fetal deaths from 0.20 to 0.18 per 100,000 ($p < 0.05$), and birth rates remain essentially unchanged. The first-stage F-statistic increases slightly to 87.86, indicating the correction does not weaken the instrument. These results confirm that our main findings are not driven by differential trends along observable dimensions correlated with baseline coal dependence.

5.2 Dynamic Adjustment and Serial Correlation

The gas-to-coal ratio trends upward nearly monotonically during our sample period, creating high serial correlation in the shift component of our instrument. Our main specification already uses the one-period lagged gas-to-coal ratio (g_{t-1}) to account for delayed employment responses to energy market shifts. However, Jaeger et al. (2018) show that when shifts are serially correlated, standard shift-share estimates may conflate contemporaneous causal

Table 12: Coal Mining Shares and Deaths of Despair with Incomplete Shares Correction

	Main Specification				Baseline Characteristics $\times$ Year FEs			
	(1) Despair	(2) Poisoning	(3) Fetal	(4) Birth	(5) Despair	(6) Poisoning	(7) Fetal	(8) Birth
Mining Employment Share	1.1639*** (0.4368)	1.0598*** (0.4081)	0.1999** (0.0859)	0.0822*** (0.0139)	0.9341** (0.3687)	0.8660** (0.3469)	0.1830** (0.0794)	0.0851*** (0.0145)
Mean Dep. Var.	45.5904	21.6477	7.1265	10.9535	45.5904	21.6477	7.1265	10.9535
First-stage F-stat	85.56	85.56	85.56	85.56	87.86	87.86	87.86	87.86
Observations	5,499	5,499	5,499	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Chars $\times$ Year FEs	No	No	No	No	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Odd-numbered columns report the main specification. Even-numbered columns add interactions of 2009 baseline county unemployment rate, poverty rate, and percent white with year fixed effects to address the incomplete shares concern in single-shift designs (Borusyak et al., 2025). Deaths of despair and poisoning rates per 100,000 total population. Fetal death rate per 100,000 total population. Birth rate per 1,000 total population. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

effects with ongoing adjustments to past shocks. Following Borusyak et al. (2025), we address this concern by controlling for the twice-lagged shift-share instrument (sig_{t-2}) in the second stage. This residualizes the current instrument on earlier lags, so identification comes from the unpredictable year-to-year innovation in the gas-to-coal ratio rather than its trending level.

Table 13 compares our main specification (columns 1-3) to the specification controlling for the twice-lagged instrument (columns 4-6). Point estimates increase substantially in that deaths of despair rise from 1.16 to 2.04 per 100,000, poisoning deaths from 1.06 to 2.34 per 100,000, and fetal deaths from 0.20 to 0.30 per 100,000. Standard errors roughly triple due to high serial correlation reducing identifying variation, but poisoning deaths remain statistically significant. The pattern of larger coefficients indicates that our static specification is conservative. The contemporaneous effect of mining employment changes on overdose deaths is larger than our main estimates suggest, but we lose precision on several key outcomes.

Table 13: Dynamic Adjustment: Controlling for Lagged Instrument

	Main Specification			Controlling for Lagged Instrument		
	(1) Despair	(2) Poisoning	(3) Fetal	(4) Despair	(5) Poisoning	(6) Fetal
Mining Employment Share	1.1639*** (0.4368)	1.0598*** (0.4081)	0.1999** (0.0859)	2.0410 (1.3506)	2.3375** (1.0613)	0.2981 (0.4817)
Mean Dep. Var.	45.5904	21.6477	7.1265	46.2004	22.0062	7.0755
First-stage F-stat	85.56	85.56	85.56	91.12	91.12	91.12
Observations	5,499	5,499	5,499	5,076	5,076	5,076
County FEs	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes
Lagged Instrument	No	No	No	Yes	Yes	Yes

Notes: 2SLS estimates. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Columns (4)–(6) add the one-year lagged shift-share instrument as a control to address concerns about serially correlated shocks and dynamic adjustment following Jaeger et al. (2018). Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper investigates how changes in coal mining employment impact both workers, their families, and the broader community. Specifically, we look at how losses due to the fracking boom in the late 2000s impacted deaths of despair and birth outcomes for the people of Appalachia. Most associate job loss with higher levels of deaths of despair for workers (Pierce and Schott, 2020). However, a working paper suggests this may not be true for Appalachian coal mine workers (Metcalf and Wang, 2019). We update this work to include the most recent literature on shift-share estimations laid out by Borusyak et al. (2025) and extend the analysis to include non-workers and birth outcomes.

Our results reveal a fundamental tradeoff in Appalachian coal communities mining employment harms workers through occupational opioid exposure while its decline harms families and vulnerable populations through economic deterioration. We find that a one percentage point increase in coal mining employment in a county increases deaths of despair by 2.5%. This is driven almost exclusively by poisoning deaths of working-age men. Mining shares do not impact deaths of despair for women. However, different from prime aged men, when coal mining shares decrease, poisoning deaths increase for children and the elderly.

This suggests a paradoxical tradeoff in that mining employment harms workers through occupational exposure to opioids, while mining job losses harm vulnerable non-workers through economic hardship and community deterioration.

Beyond deaths of despair, higher mining shares lead to more births, but those children are more likely to be born preterm and at a low birth weight. Similarly, higher levels of mining employment are associated with increased fetal deaths that are concentrated late in pregnancy, which is often affiliated with opioid use by mothers. Taken together, these suggest that even though mothers are not more likely to die from overdoses they experience negative impacts from coal mining.

Finally, we show that this is an Appalachian phenomenon. Utilizing coal mining shares across the United states, we show that coal mining shares do not predict deaths of despair outside of Appalachia. This highlights the unique setting in Appalachia due to its heightened exposure to the opioid epidemic combined with its relative isolation.

Policymakers have suggested two pathways with one side wanting to bring coal jobs back to Appalachia and the other wanting to retrain former miners into other occupations. These results suggest that restoring coal employment without addressing occupational health risks may perpetuate the opioid mortality burden on workers. Retraining programs for displaced miners must account for the health consequences borne by workers' families and communities during the transition period. Moreover, if the occupational exposure mechanism generalizes beyond mining, retraining workers into other physically demanding industries may replicate rather than resolve the opioid mortality risk.

This study demonstrates that the health consequences of industry decline depend critically on the nature of the jobs being lost. Unlike manufacturing displacement, which harms workers through economic distress, coal mining job losses reveal a protective effect by reducing occupational exposure to prescription opioids. However, this aggregate benefit masks severe costs for families and communities left behind. Energy transitions and labor market policies must account for these competing effects, balancing worker safety against household economic

security. Future work should focus on the role of migration in these patterns. Work by Krause (2023) suggests that selective migration may attenuate these results for those left behind. It may also be useful to compare to other hazardous working environments that often take place in isolated areas such as logging.

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A Appendix Tables

A.1 Progressive Specification

Table A.1: Coal Mining Employment Share and Deaths of Despair

	Deaths of Despair (per 100k)		
	(1)	(2)	(3)
Mining Employment Share	0.8759** (0.4322)	1.1539*** (0.4347)	1.1639*** (0.4368)
First-stage F-stat	81.78	85.15	85.56
Observations	5,499	5,499	5,499
County FEs	Yes	Yes	Yes
Year FEs	Yes	No	Yes
State $\times$ Year FEs	No	Yes	Yes
Demographic Controls	No	No	Yes

Notes: 2SLS estimates. Mining employment share measured in percentage points. Column (1) includes county and year fixed effects. Column (2) replaces year FEs with state $\times$ year FEs. Column (3) adds demographic controls. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Robustness to Population Controls

Table A.2: Robustness: Count of Coal Mining Employment

	(1)	(2)	(3)	(4)	(5)
	Despair	Poisoning	Fetal	Births	Opioids
Mining Employment Share	0.6084** (0.2692)	0.5578** (0.2438)	0.0600** (0.0235)	3.9546*** (1.2109)	108.6799 (209.7824)
Mean Dep. Var.	26.4173	13.0962	4.4079	679.6401	37,768.80
First-stage F-stat	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,115
County FEs	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Dependent variables are raw counts. Mining employment share measured in percentage points. Opioid counts measured in grams of active ingredient (ARCOS). No population controls included; see text for discussion. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Robustness: Population-Weighted Estimates

	(1)	(2)	(3)	(4)	(5)
	Despair	Poisoning	Fetal	Births	Opioids
Mining Employment Share	1.4860*** (0.5092)	1.4119*** (0.4951)	0.1824*** (0.0690)	0.0610*** (0.0128)	371.4912 (650.0313)
Mean Dep. Var.	45.5904	21.6477	7.1265	10.9535	61,540.69
First-stage F-stat	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,115
County FEs	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates weighted by total county population. Mining employment share measured in percentage points. Despair and poisoning rates per 100,000 total population. Fetal death rate per 100,000 total population. Birth rate per 1,000 total population. Opioid rate per 100,000 total population in grams (ARCOS). Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Additional Heterogeneity by Age

Table A.4: Coal Mining Employment Share and Suicide Deaths by Age Group

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	0–17	18–24	25–34	35–44	45–54	55–64	65+
Mining Employment Share	0.0012 (0.0920)	0.0115 (0.0700)	0.0177 (0.0898)	−0.0632 (0.1069)	0.0495 (0.0645)	−0.1770** (0.0807)	0.6271 (0.4976)
Mean Dep. Var.	2.0878	2.3091	4.1968	5.0003	5.7805	4.7857	20.5468
First-stage F-stat	85.56	85.56	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Rates per 100,000 age-group-specific population. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Coal Mining Employment Share and Alcohol-Related Deaths by Age Group

	(1)	(2)	(3)	(4)	(5)	(6)
	18–24	25–34	35–44	45–54	55–64	65+
Mining Employment Share	−0.0003 (0.0004)	−0.0138 (0.0221)	−0.0211 (0.0504)	0.0473 (0.0804)	0.2000** (0.0870)	−0.1723 (0.2518)
Mean Dep. Var.	0.0076	0.3078	1.2532	3.9060	4.5849	10.3112
First-stage F-stat	85.56	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Rates per 100,000 age-group-specific population. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Ages 0–17 excluded due to insufficient variation. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Coal Mining Employment Share and Fetal Deaths by Mother's Age

	(1)	(2)	(3)	(4)
	<18	18–24	25–34	35–44
Mining Employment Share	0.0041 (0.0113)	0.0720* (0.0371)	0.0832 (0.0741)	0.0361** (0.0183)
Mean Dep. Var.	0.2342	2.6166	3.2919	0.9625
First-stage F-stat	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. Fetal death rates per 100,000 total population. Mining employment share measured in percentage points. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Female Population Denominator

Table A.7: Robustness: Defining Birth Outcomes with Female Population Denominator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Fetal Total	Fetal Early	Fetal Late	Births	LBW	VLBW	Preterm
Mining Employment Share	0.4027** (0.1800)	0.1679 (0.1373)	0.2448** (0.1040)	0.1545*** (0.0250)	1.9674** (0.8793)	0.1356 (0.1974)	2.9750* (1.7868)
Mean Dep. Var.	14.0878	6.3247	7.6712	21.6782	194.2819	32.0092	281.6882
First-stage F-stat	85.56	85.56	85.56	85.56	85.56	85.56	85.56
Observations	5,499	5,499	5,499	5,499	5,499	5,499	5,499
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 2SLS estimates. All rates use female population as the denominator. Fetal death rates per 100,000 female population. Birth rate per 1,000 female population. LBW, VLBW, and preterm rates per 100,000 female population. Mining employment share measured in percentage points; coefficients represent the effect of a one percentage point increase. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.